VII. Review of non-linear continuum mechanics

1. Kinematics of large deformations

2. Balance laws

3. Constitutive equations

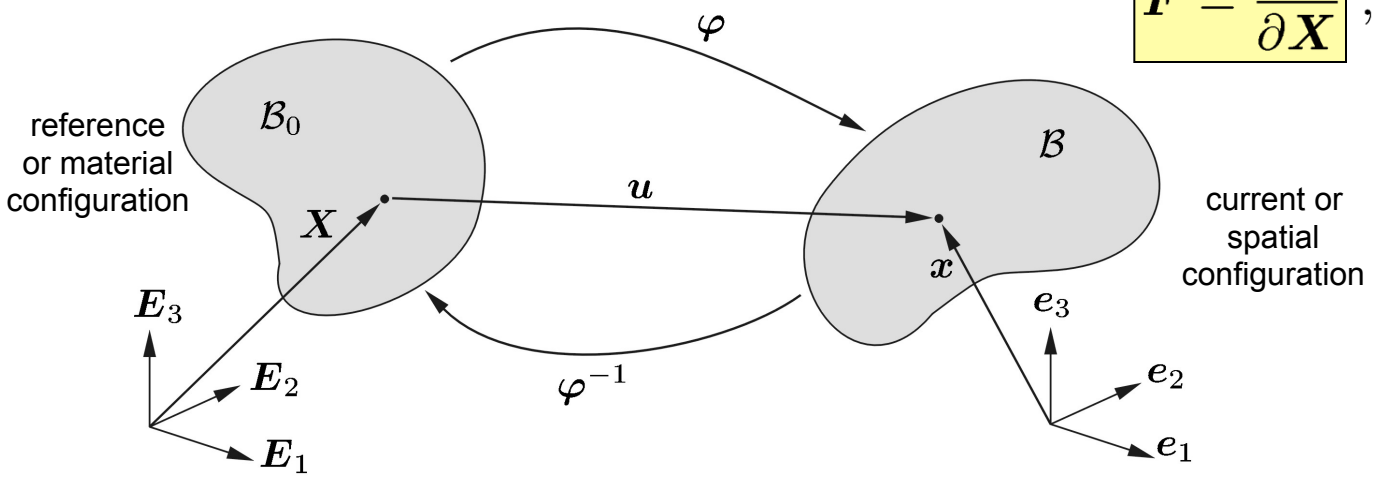
4. Mechanical BVP

5. Variational Principles

Large deformation kinematics

Deformation

$$\boldsymbol{x} = \boldsymbol{\varphi}(\boldsymbol{X})$$



Deformation gradient

$$\boxed{\boldsymbol{F} = \frac{\partial \boldsymbol{\varphi}}{\partial \boldsymbol{X}}} \,, \quad F_{iA} = \frac{\partial \varphi_i}{\partial X_A}$$

Changes of volumes, areas, lengths

$$\mathrm{d}v = J\,\mathrm{d}V \;, \quad J = \det \boldsymbol{F}$$

$$\boldsymbol{n}\,\mathrm{d}a = J\,\boldsymbol{F}^{-T}\,\boldsymbol{N}\,\mathrm{d}A$$

$$\mathrm{d}s = \sqrt{\boldsymbol{M}\cdot\boldsymbol{C}\boldsymbol{M}}\,\mathrm{d}S \;, \quad \boldsymbol{C} = \boldsymbol{F}^T\boldsymbol{F}$$

Displacement

$$\boldsymbol{u}(\boldsymbol{X}) = \boldsymbol{x}(\boldsymbol{X}) - \boldsymbol{X}$$

Displacement gradient

$$\boldsymbol{H} = \frac{\partial \boldsymbol{u}}{\partial \boldsymbol{X}} = \boldsymbol{F} - \boldsymbol{1}$$

Stretch and strain measures

Stretch tensors $\boldsymbol{U}$, $\boldsymbol{V}$, principal stretches λ_A and directions $\boldsymbol{N}_A$, $\boldsymbol{n}_A$

$$\boldsymbol{C} = \boldsymbol{F}^T \boldsymbol{F} \qquad \boldsymbol{U} = \sqrt{\boldsymbol{C}} \qquad \boldsymbol{U} = \sum_{A=1}^{3} \lambda_A \, \boldsymbol{N}_A \otimes \boldsymbol{N}_A$$

$$\boldsymbol{B} = \boldsymbol{F} \boldsymbol{F}^T \qquad \boldsymbol{V} = \sqrt{\boldsymbol{B}} \qquad \boldsymbol{V} = \sum_{A=1}^{3} \lambda_A \, \boldsymbol{n}_A \otimes \boldsymbol{n}_A$$

Polar decomposition

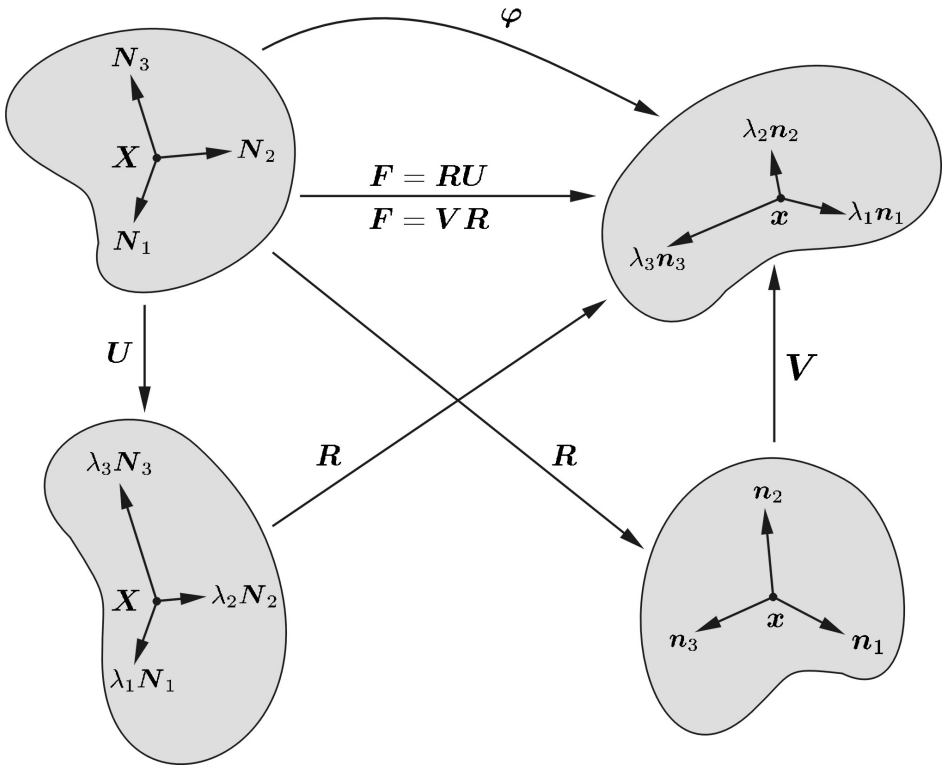
$$\boxed{\boldsymbol{F} = \boldsymbol{R}\boldsymbol{U} = \boldsymbol{V}\boldsymbol{R}}$$

$$\boldsymbol{R} = \sum_{A=1}^{3} \boldsymbol{n}_A \otimes \boldsymbol{N}_A$$

Strain measures

$$\boldsymbol{E} = \tfrac{1}{2}(\boldsymbol{C} - \boldsymbol{1})$$

$$\boldsymbol{e} = \tfrac{1}{2}(\boldsymbol{1} - \boldsymbol{B}^{-1})$$



Momentum balance

Balance of linear momentum

$$\dot{\boldsymbol{p}} = \boldsymbol{F}_{\text{ext}} \qquad \text{with} \quad \boldsymbol{p} = \int_{\mathcal{P}} \rho\, \boldsymbol{v}\, \mathrm{d}v\ , \qquad \boldsymbol{F}_{\text{ext}} = \int_{\mathcal{P}} \rho\, \boldsymbol{b}\, \mathrm{d}v + \int_{\partial\mathcal{P}} \boldsymbol{t}\, \mathrm{d}a$$

$$\Longleftrightarrow \quad \boxed{\operatorname{div} \boldsymbol{\sigma} + \rho\, \boldsymbol{b} = \rho\, \boldsymbol{a}}$$

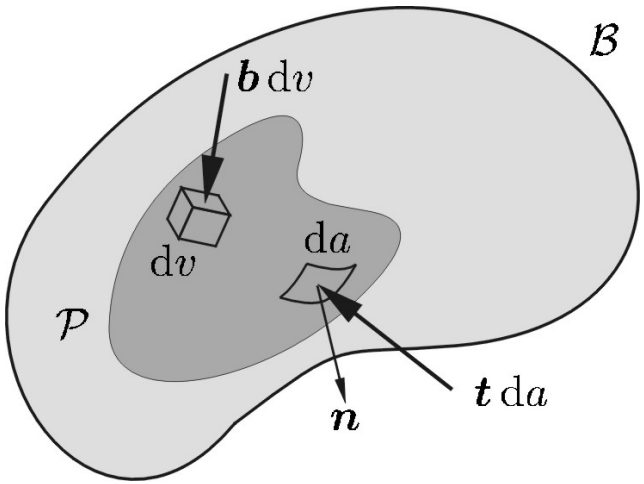
Cauchy formula

$$\boldsymbol{t} = \boldsymbol{\sigma}\boldsymbol{n}$$

Balance of angular momentum w.r.t. origin

$$\dot{\boldsymbol{L}} = \boldsymbol{M}_{\text{ext}} \qquad \boldsymbol{M}^{(0)}_{\text{ext}} = \int_{\mathcal{P}} \rho\, \boldsymbol{x} \times \boldsymbol{b}\, \mathrm{d}v + \int_{\partial\mathcal{P}} \boldsymbol{x} \times \boldsymbol{t}\, \mathrm{d}a$$

$$\Longleftrightarrow \quad \boxed{\boldsymbol{\sigma} = \boldsymbol{\sigma}^T} \qquad \boldsymbol{L}^{(0)} = \int_{\mathcal{P}} \rho\, \boldsymbol{x} \times \boldsymbol{v}\, \mathrm{d}v$$



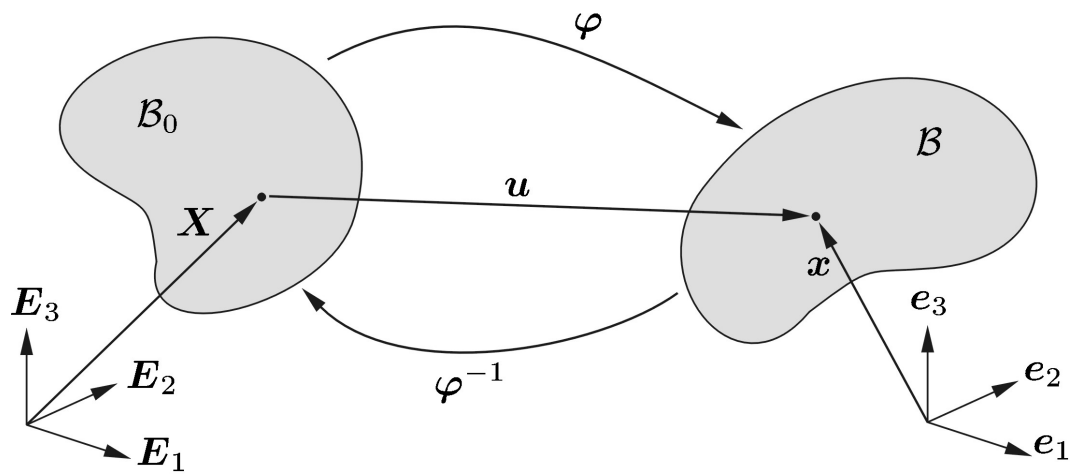
Stress tensors

	Kirchhoff stress	$\boldsymbol{\tau}$	$= J\boldsymbol{\sigma}$
1.	Piola Kirchhoff stress	$\boldsymbol{P}$	$= \boldsymbol{\tau}\boldsymbol{F}^{-T}$
2.	Piola Kirchhoff stress	$\boldsymbol{S}$	$= \boldsymbol{F}^{-1}\boldsymbol{P}$

Constitution: finite elasticity

Cauchy elasticity

$$\boldsymbol{\sigma} = \boldsymbol{\sigma}\big(\boldsymbol{F}(\boldsymbol{X},t),\boldsymbol{X}\big)$$



Hyperelasticity

$$W = W(\boldsymbol{F},\boldsymbol{X})$$

$$\boxed{\boldsymbol{P} = \frac{\partial W}{\partial \boldsymbol{F}}}$$

$$\boldsymbol{S} = \frac{\partial W}{\partial \boldsymbol{E}} = 2\frac{\partial W}{\partial \boldsymbol{C}}$$

Material tangent

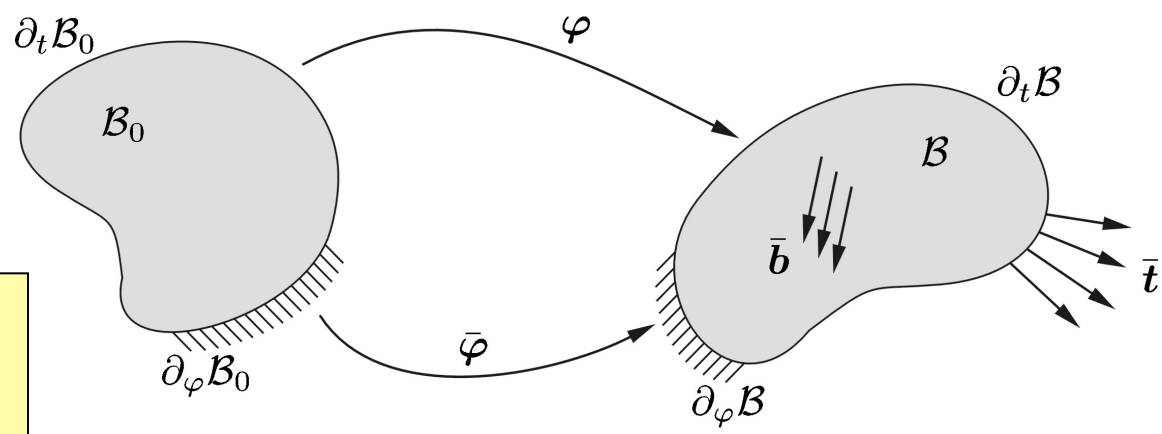
$$\mathbb{C} = \frac{\partial \boldsymbol{S}}{\partial \boldsymbol{E}} = \frac{\partial^2 W}{\partial \boldsymbol{E}\,\partial \boldsymbol{E}}$$

Special cases

isotropy: $W = W(I_1, I_2, I_3)$

Neo-Hooke: $W(I_1, I_3) = \frac{\Lambda}{4}\big(J^2 - 1 - 2\ln J\big) + \frac{\mu}{2}\big(I_1 - 3\big) - \mu \ln J$

Mechanical BVP: strong form



Spatial description

Given: $\mathcal{B}_0,\ \bar{\boldsymbol{\varphi}},\ \bar{\boldsymbol{b}},\ \bar{\boldsymbol{t}},\ W$

Find: $\boldsymbol{\varphi}(\boldsymbol{X}) \quad \forall\, \boldsymbol{X} \in \mathcal{B}_0$

satisfying

$$\boxed{\begin{aligned} \operatorname{div} \boldsymbol{\sigma} + \rho \bar{\boldsymbol{b}} &= \boldsymbol{0} && \text{in } \mathcal{B} \\ \boldsymbol{\varphi} &= \bar{\boldsymbol{\varphi}} && \text{on } \partial_\varphi \mathcal{B} \\ \boldsymbol{t} = \boldsymbol{\sigma}\boldsymbol{n} &= \bar{\boldsymbol{t}} && \text{on } \partial_t \mathcal{B} \end{aligned}}$$

with

$$\boldsymbol{F} = \frac{\partial \boldsymbol{\varphi}}{\partial \boldsymbol{X}}\,, \qquad \boldsymbol{P} = \frac{\partial W}{\partial \boldsymbol{F}}$$

$$\boldsymbol{\sigma} = \boldsymbol{P}\boldsymbol{F}^T / J$$

Alternative material description

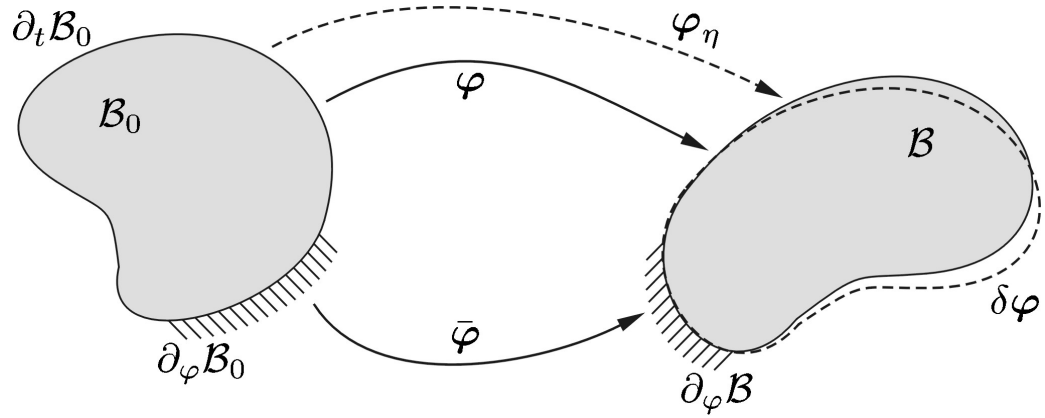
$$\boxed{\begin{aligned} \operatorname{Div} \boldsymbol{P} + \rho_0 \bar{\boldsymbol{B}} &= \boldsymbol{0} && \text{in } \mathcal{B}_0 \\ \boldsymbol{\varphi} &= \bar{\boldsymbol{\varphi}} && \text{on } \partial_\varphi \mathcal{B}_0 \\ \boldsymbol{T} = \boldsymbol{P}\boldsymbol{N} &= \bar{\boldsymbol{T}} && \text{on } \partial_t \mathcal{B}_0 \end{aligned}}$$

Variational principles

Variation of the deformation

$$\boldsymbol{\varphi}_\eta := \boldsymbol{\varphi} + \eta\,\delta\boldsymbol{\varphi}\ , \quad \eta \in \mathbb{R}$$

$$\mathcal{V} = \left\{ \boldsymbol{\varphi} : \mathcal{B}_0 \to \mathbb{R}^3 \,\middle|\, \boldsymbol{\varphi} = \mathbf{0} \right\}$$



The weak form: *the principle of virtual work (PVW)*

spatial description
$$\int_{\mathcal{B}} \boldsymbol{\sigma} : \operatorname{grad}(\delta\boldsymbol{\varphi})\,\mathrm{d}v = \int_{\mathcal{B}} \delta\boldsymbol{\varphi}\cdot\rho\,\bar{\boldsymbol{b}}\,\mathrm{d}v + \int_{\partial_t\mathcal{B}} \delta\boldsymbol{\varphi}\cdot\bar{\boldsymbol{t}}\,\mathrm{d}a \qquad \forall\ \delta\boldsymbol{\varphi} \in \mathcal{V}$$

material description
$$\int_{\mathcal{B}_0} \boldsymbol{P} : \operatorname{Grad}(\delta\boldsymbol{\varphi})\,\mathrm{d}V = \int_{\mathcal{B}_0} \delta\boldsymbol{\varphi}\cdot\rho_0\bar{\boldsymbol{B}}\,\mathrm{d}V + \int_{\partial_t\mathcal{B}_0} \delta\boldsymbol{\varphi}\cdot\bar{\boldsymbol{T}}\,\mathrm{d}A$$

Variational principles: *the principle of stationary potential energy*

$$\Pi(\boldsymbol{\varphi}) = \int_{\mathcal{B}_0} W(\boldsymbol{F})\,\mathrm{d}V - \Pi_{\mathrm{ext}}(\boldsymbol{\varphi})$$

$$\delta\Pi = 0 \quad \Rightarrow \quad \text{PVW}$$

example

$$\Pi_{\mathrm{ext}}(\boldsymbol{\varphi}) = \int_{\mathcal{B}_0} \boldsymbol{\varphi}\cdot\rho_0\bar{\boldsymbol{B}}\,\mathrm{d}V + \int_{\partial_t\mathcal{B}_0} \boldsymbol{\varphi}\cdot\bar{\boldsymbol{T}}\,\mathrm{d}A$$